

Anastasiia Mokhonko

DATA SCIENCE & AI STUDENT

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“Aspiring Data Scientist and AI Specialist passionate about digital innovation, machine learning, and solving real-world problems.”

Summary

I'm a machine learning engineer focused on building systems that actually work in production. I've delivered projects for Content Intelligence Agency, NPEC plant research, NAC Breda Football Club, and ANWB. My work spans NLP, computer vision, and MLOps. I use Python, Azure ML, and frameworks like BERT, CamemBERT, and U-Net to solve real problems. Recent results include 95% accuracy in disease detection, 99% faster analysis workflows, and systems handling 44 requests per second. I handle everything from data processing and model training to deployment with Docker, CI/CD pipelines, and monitoring. I also work with AI ethics requirements like GDPR and EU AI Act compliance.

Experience

Emotion Detection NLP Pipeline – Content Intelligence Agency (Phase 2)

Breda, NL

MLOPS ENGINEER, DEVELOPER

May 2025 – Jun 2025

- Deployed production-ready emotion detection system classifying 6 emotions in French video/audio content with 44.7 requests/second capacity.
- Built Azure ML pipelines with automated training, retraining, and deployment using Docker containers and CI/CD integration.
- Developed secure FastAPI endpoints and Streamlit interface for multi-user access with monitoring and logging.
- Implemented blue-green deployment strategy with model versioning and rollback capabilities for zero-downtime updates.

Emotion Detection from Video – Content Intelligence Agency (Phase 1)

Breda, NL

DEVELOPER, NLP ENGINEER

Feb 2025 – Apr 2025

- Developed NLP pipeline for emotion analysis in French reality TV including audio transcription, translation, and classification.
- Implemented and compared Naive Bayes, Logistic Regression, LSTM, BERT, and CamemBERT models for emotion detection.
- Applied XAI techniques (LIME, SHAP) for model interpretability; delivered research documentation and model card to client.

Root Segmentation & Robotic Control – NPEC Netherlands Plant Eco-phenotyping Centre

Breda, NL

DEVELOPER, COMPUTER VISION ENGINEER

Nov 2024 – Dec 2024

- Built computer vision pipeline using U-Net for plant root segmentation and graph theory (NetworkX, Dijkstra) for measurement.
- Developed PID and Reinforcement Learning controllers (Stable Baselines 3) for OpenTrons OT-2 robotic microbial application.
- Achieved 99% time reduction in root analysis workflow; delivered high-accuracy solution to research client.

Cybersecurity Awareness in Small Hotels – Policy Research for DigiWerkplaats

Breda, NL

ANALYTICS TRANSLATOR, TEAM LEAD, RESEARCHER

Sep 2024 – Oct 2024

- Led research team of 5 investigating employee cybersecurity awareness impact on hotel security using mixed-methods approach.
- Designed and analyzed surveys (Qualtrics) and interviews; ensured GDPR compliance and research ethics approval.
- Found moderate positive correlation ($r=0.456$) between training frequency and security adherence; recommended quarterly training.

ANWB Safe Driving Application – ANWB Dutch Automobile Association

Breda, NL

DEVELOPER, DATA ANALYST

May 2024 – Jun 2024

- Developed Streamlit application analyzing driver behavior and road safety using ANWB datasets and time series analysis.
- Implemented incident classification models with EU AI Act compliance and Agile project management methodology.

Farmer's Buddy: Potato Blight Detection AI

Breda, NL

DEVELOPER, DATA SCIENTIST

Feb 2024 – Apr 2024

- Built AI-driven mobile application for potato blight detection using MobileNetV2 transfer learning, achieving 95.3% accuracy.
- Applied Grad-CAM for explainable AI and conducted think-aloud usability studies to optimize user experience.

Goal Prediction Analytics – NAC Breda Football Club

Breda, NL

DATA ANALYST, MACHINE LEARNING ENGINEER

Nov 2023 – Jan 2024

- Developed machine learning model analyzing 16,535 players to predict goal-scoring potential with 95%+ accuracy.
- Delivered GDPR-compliant recruitment recommendations using linear regression and correlation analysis.

Human Trafficking Data Analysis Dashboard – UN SDG 16

Breda, NL

DEVELOPER, DATA ANALYST

Sep 2023 – Nov 2023

- Designed interactive Power BI dashboard analyzing global trafficking patterns by age, gender, exploitation type, and region.
- Calculated temporal correlation ($r=0.68$) and delivered policy recommendations for prevention strategies supporting SDG 16.

Education

ITSTEP Academy

PYTHON DEVELOPER COURSE

Kyiv, Ukraine

2022 – 2023

- Completed 6-month intensive Python programming and backend development course.
- Covered Python fundamentals, OOP, data structures, algorithms, SQL/NoSQL databases, and Django web development.
- Gained hands-on experience in asynchronous programming, software testing, Git version control, and cloud technologies.

Extracurricular Activity

Degree Programme Committee (DPC) – Data Science & AI Programme

Breda University of Applied Sciences, Breda, NL

2024 – Present

- STUDENT REPRESENTATIVE
- Selected as one of three student representatives to serve on the official programme committee alongside faculty members.
 - Collaborate with lecturers and programme management to improve curriculum quality, course delivery, and student experience.
 - Propose ideas for programme enhancement, address student concerns, and help resolve issues encountered by both students and staff.
 - Participate in regular committee meetings to review programme policies, assessment methods, and educational standards.
 - Act as liaison between student body and faculty, ensuring student perspectives are represented in programme decisions.

Nakormim Bali – Orphanage Support Organization

Bali, Indonesia

2022 – Present

- VOLUNTEER & MONTHLY DONOR
- Support post-COVID-19 pandemic relief efforts for orphanages, helping to feed children and facilitate adoption processes to find them forever homes.
 - Provide ongoing monthly financial donations to sustain orphanage operations and child welfare programs.
 - Organized educational study sessions for children, creating supportive learning environments and mentoring opportunities.
 - Coordinated group excursions including beach trips, turtle hatching observation events, and creative activities such as pottery workshops.
 - Contribute to holistic child development through recreational, educational, and cultural experiences.

Technical Skills

Programming	Python, SQL, HTML/CSS, LaTeX
ML/AI Frameworks	PyTorch, TensorFlow, Scikit-learn, Transformers (HuggingFace), BERT, CamemBERT, U-Net
MLOps & Cloud	Azure ML, Docker, CI/CD, Git, FastAPI, Airflow
Data & Visualization	Power BI, Pandas, NumPy, Matplotlib, Seaborn, Streamlit, SQL (PostgreSQL, MySQL)
Specializations	NLP, Computer Vision, Deep Learning, Time Series Analysis, XAI (LIME, SHAP)
Tools & Platforms	VSCode, Jupyter, Git/GitHub, Azure DevOps, Figma, Trello

Languages

Native	Russian, Ukrainian
Professional	English (C1 - Advanced)
Intermediate	French (B2 - Upper Intermediate)